**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 136 · Lesson 136 · Normalisation: First and Second Normal Form

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What table resolves a many-to-many database relationship?

Q2. What is a composite key?

Q3. When does a linear search stop?

Q4. In Big O notation, what is binary search worst-case complexity?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

### Key vocabulary

1NF — Attributes hold atomic values and rows can be uniquely identified.

2NF — In 1NF, no non-key attribute depends on only part of a composite candidate key.

partial dependency — Non-key field depends on a subset of a composite key.

anomaly — Insertion, update or deletion causes undesirable effects.

### New learning

Make fields atomic — Replace 'Maths, Art' in one cell with separate enrolment records.

Identify keys — Check whether key is composite, such as StudentID+CourseID.

Remove partial dependencies — Move StudentName to Student and CourseTitle to Course.

### Worked example

Enrolment(StudentID,CourseID,StudentName,CourseName,Grade) has composite PK (StudentID,CourseID). Achieve 2NF.

1. StudentID+CourseID key — Find the given data and name the required result.

2. Names partly depend on key — Explain why this intermediate stage is necessary.

3. Split Student and Course — StudentName depends only on StudentID; CourseName only on CourseID. Split into Student(StudentID PK,StudentName), Course(CourseID PK,CourseName), Enrolment(StudentID FK,CourseID FK,Grade) with composite PK.

# We Do · Practise with guidance

OrderLine(OrderID,ProductID,ProductName,Qty) uses composite PK. Identify partial dependency and fix.

Composite OrderID+ProductID

Find the given data and name the required result.

ProductName depends on ProductID

Explain why this intermediate stage is necessary.

Move name to Product

ProductName depends on ProductID alone, not whole OrderID+ProductID key. Create Product(ProductID PK,ProductName) and retain OrderLine(OrderID,ProductID,Qty).

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Explain atomicity in 1NF and correct a cell containing 'Red, Blue, Green'. [6]

### Question 2

Explain partial dependency using OrderID+ProductID key and ProductDescription. [6]

### Question 3

Convert StudentID,ClubID,StudentName,ClubName,JoinDate with composite key to 2NF. [8]

# Exit Ticket · Answer before checking

A relation with a single-attribute candidate key can have a partial dependency on that key.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. A junction/associative table.

Q2. A key made from two or more attributes together.

Q3. On a match or after examining all items.

Q4. O(log n) for sorted random-access data.

### You Do pages · Independent questions

Q1. Atomic field requirement [2], problem identified [2], separate rows or related table [2].

Q2. Composite key identified [2], description depends on ProductID only [2], violates 2NF and should move [2].

Q3. Student table [2], Club table [2], Membership table with composite key/date [2], correct FK/dependency reasoning [2].

### Exit Ticket · Model answer

False. Partial-key dependency requires a composite candidate key.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**